

Cube and cube root

* Exceptional Cases:

$$\sqrt{1296} = 34 \times$$

$$= 36 \checkmark$$

$$32^2 = 1024$$

* Hatke Cases:

$$\sqrt{961} = 31$$

$$\sqrt{1681} = 41$$

* Perfect Square

$$\sqrt{19341} \Rightarrow 139$$

$$\sqrt{19321} \Rightarrow 139 \checkmark$$

2, 3, 7, 8 $\Rightarrow$ Unit digit

(Last place) $\rightarrow$ No square root.

* Unit digit 5:

Upper (compulsory)

$$\sqrt{\textcircled{25}} \rightarrow 5$$

$$\sqrt{9025}$$

$$\rightarrow 95$$

$$\sqrt{4225}$$

$$65$$

$$\sqrt{30625}$$

$$175$$

$$\sqrt{60025}$$

$$245$$

* Unit digit 0:-

$$\sqrt{14400} \rightarrow 120$$

$$\sqrt{900} \rightarrow 30$$

$$\text{eg: } \sqrt{4900} = 70$$

$$\sqrt{16900} = 130$$

$$\sqrt{52900} = 170$$

$$\sqrt{72900} = 270$$

* Cubes

$$1^3 = 1$$

$$7^3 = 343$$

$$13^3 = 2197$$

$$2^3 = 8$$

$$8^3 = 512$$

$$14^3 = 2744$$

$$3^3 = 27$$

$$9^3 = 729$$

$$15^3 = 3375$$

$$4^3 = 64$$

$$10^3 = 1000$$

$$16^3 = 4096$$

$$5^3 = 125$$

$$11^3 = 1331$$

$$6^3 = 216$$

$$12^3 = 1728$$

Ex-1) $\sqrt[3]{238328}$
 $\downarrow$
 3280488
 perfect cube
 $\downarrow$
 62

Unit digit

Unit digit Possibility

$$1 \rightarrow 1$$

$$2 \rightarrow 8 *$$

$$3 \rightarrow 7 *$$

$$4 \rightarrow 4$$

$$5 \rightarrow 5$$

$$6 \rightarrow 6$$

$$7 \rightarrow 3 *$$

$$8 \rightarrow 2 *$$

$$9 \rightarrow 9$$

$$0 \rightarrow 0$$

2) $\sqrt[3]{140608}$
 52

3) $\sqrt[3]{2299968}$
 $\downarrow$
 132

2 3 7 8

$$\sqrt[3]{857375}$$

95

$$\sqrt[3]{59319}$$

39

$$\sqrt[3]{389017}$$

73

$$\sqrt[3]{185193}$$

53

$$\sqrt[3]{54872}$$

38

$$\sqrt[3]{2000376}$$

126

$$\begin{array}{r} 3-7 \\ 7-3 \end{array} \quad \begin{array}{r} 2-8 \\ 8-2 \end{array}$$

$$\sqrt[3]{21952} \\ 28$$

$$\sqrt[3]{125000} \\ 50$$

$$\sqrt[3]{2744} \\ 14$$

$$\sqrt[3]{46656} \\ 36$$

$$\sqrt[3]{195112} \\ 58$$

$$\sqrt[3]{1367631} \\ 111$$

$$\sqrt[3]{357911} \\ 71$$

$$\sqrt[3]{74088} \\ 42$$

$$\sqrt[3]{2248091} \\ 131$$

$$\sqrt[3]{250047} \\ 63$$

$$\sqrt[3]{3307949} \\ 149$$

$$\sqrt[3]{35937} \\ 33$$

* Approximation (Nearest)

$$\text{eg- } \sqrt{730} = \sqrt{729} \\ \approx 27$$

$$\sqrt[3]{730} = \sqrt[3]{729} \\ = 9$$

$$\sqrt{575} \approx 24$$

$$\sqrt[3]{220} \approx 6$$

$$\sqrt{125} \approx 11$$

$$\sqrt[3]{121} \approx 5$$

$$\sqrt{200} \approx 14$$

$$\sqrt[3]{1333} \approx 11$$

$$\sqrt{512} \approx 22$$

$$\sqrt[3]{4003} \approx 16$$

* Home Work:

①-

$$\sqrt[3]{1404928} \\ 112$$

$$\sqrt[3]{3652264} \\ 154$$

$$\sqrt[3]{175616} \\ 56$$

$$\sqrt[3]{2097152} \\ 128$$

$$\sqrt[3]{474552} \\ 78$$

$$\sqrt[3]{117649} \\ 49$$

$$\sqrt[3]{571787} \\ 83$$

$$\sqrt[3]{373248} \\ 72$$

$$\sqrt[3]{1404928} \\ 112$$

$$\sqrt[3]{2571353} \\ 137$$

$$\sqrt[3]{456533} \\ 77$$

$$\sqrt[3]{884736} \\ 96$$

$$\sqrt[3]{46656} \\ 36$$

$$\sqrt[3]{195112} \\ 58$$

$$\sqrt[3]{970299} \\ 99$$

$$\sqrt[3]{35937} \\ 33$$

$$\sqrt[3]{2299968} \\ 132$$

$$\sqrt[3]{21952} \\ 28$$

$$\textcircled{2} - \sqrt[3]{250047} \\ 63$$

$$\sqrt[3]{857375} \\ 95$$

$$\sqrt[3]{1367631} \\ 111$$

$$\sqrt[3]{257911} \\ 71$$

$$\sqrt[3]{3511808} \\ 152$$

$$\sqrt[3]{2000376} \\ 126$$

$$\sqrt[3]{681472} \\ 88$$

$$\sqrt[3]{884736} \\ 96$$

$$\sqrt[3]{42875} \\ 35$$

$$\sqrt[3]{292427} \\ 143$$

$$\sqrt[3]{205379} \\ 59$$

$$\sqrt[3]{551368} \\ 82$$

$$\sqrt[3]{2248091} \\ 131 \text{ ~~121~~}$$

$$\sqrt[3]{1860867} \\ 127$$

$$\sqrt[3]{3112136} \\ 146$$

$$\sqrt[3]{166375} \\ 55$$

$$\sqrt[3]{74088} \\ 42$$

$$\sqrt[3]{17576} \\ 26$$

$$\textcircled{3} - \sqrt[3]{658503} \\ 87$$

$$\sqrt[3]{59319} \\ 39$$

$$\sqrt[3]{54872} \\ 38$$

$$\sqrt[3]{2248091} \\ 131$$

$$\sqrt[3]{884736} \\ 96$$

$$\sqrt[3]{3112136} \\ 146$$

$$\sqrt[3]{185193} = 57$$

$$\sqrt[3]{1367631} = 111$$

$$\sqrt[3]{314432} = 68$$

$$\sqrt[3]{79507} = 43$$

$$\sqrt[3]{389017} = 73$$

$$\sqrt[3]{474552} = 78$$

$$\sqrt[3]{1404928} = 112$$

$$\sqrt[3]{912673} = 97$$

$$\sqrt[3]{1860867} = 123$$

$$\sqrt[3]{405224} = 74$$

$$\sqrt[3]{3307949} = 149$$

$$\sqrt[3]{157464} = 54$$

Session - 5

11/09/2028

* Decimal - 1

$$\bullet (24)^2 = 576$$

$$(2.4)^2 = 5.76 \quad \begin{array}{l} 1 \times 2 \end{array}$$

$$(0.24)^2 = 0.0576 \quad \begin{array}{l} 2 \times 2 \end{array}$$

$$(0.024)^2 = 0.000576 \quad \begin{array}{l} 3 \times 2 \end{array}$$

} $\times 2$

$$\bullet (8)^3 = 512$$

$$(0.8)^3 = 0.512 \quad \begin{array}{l} 1 \times 3 = 3 \end{array}$$

$$(0.08)^3 = 0.000512 \quad \begin{array}{l} 2 \times 3 = 6 \end{array}$$

$$(0.008)^3 = 0.00000512 \quad \begin{array}{l} 3 \times 3 = 9 \end{array}$$

} $\times 3$

$$\bullet \sqrt{676} = 26$$

$$\sqrt{6.76} = 2.6$$

$$\sqrt{0.676} = 0.82$$

$$\sqrt{0.0676} = 0.26$$

} $\div 2$

$$\begin{aligned} \sqrt[3]{1331} &= 11 \\ \sqrt[3]{1.331} &= 1.1 \\ \sqrt[3]{0.001331} &= 0.11 \end{aligned}$$

1/3

$$\sqrt[3]{0.000004096} = 0.016$$

Eg- 1) $(0.008)^2 \rightarrow 0.000064$ 2) $(0.04)^3 \rightarrow 0.000064$

3) $\sqrt{0.000729} \rightarrow 0.027$

4) $\sqrt[3]{0.000729} \rightarrow 0.09$

• $\frac{144}{3} = 48$

• $\frac{144}{0.003} = 48000$

$\frac{14.4}{3} = 4.8$

$\frac{144}{0.3} = 480$

$\frac{0.144}{3} = 0.048$

~~$\frac{144}{0.03} = 4800$~~

• $\frac{1.44}{0.03} = 48$

$\frac{14.4}{0.0003} = 48000$

$\frac{0.00144}{0.03} = 0.048$

$N^r \Rightarrow$ Backside
 $D^r \Rightarrow$ No. of zeros

Eg-1) $\frac{(0.04)^2 \times \sqrt{0.000729}}{\sqrt[3]{0.000729} \times (0.02)^5}$

$= 0.0016 \times 0.027$

0.09×0.000008

$= 0.00072$

$= 200 \times 0.3$

$= 2 \times 0.3$

$= 0.6$

$= 0.6$

* Important Number:

- 1) Whole No $\rightarrow 0, 1, 2, 3, 4, 5, \dots \infty$
- 2) Natural No $\rightarrow 1, 2, 3, 4, 5, \dots \infty$
- 3) Even No $\rightarrow 0, 2, 4, 6, 8, 10, \dots \infty$
- 4) Odd No $\rightarrow 1, 3, 5, 7, 9, \dots \infty$
- 5) Integers $\rightarrow -\infty \dots -3, -2, -1, 0, 1, 2, 3, \dots \infty$
(-ve, 0, +ve)

6) Prime No $\rightarrow$

2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
31, 37, 41, 43, 47, 53, 59,
61, 67, 71, 73, 79, 83, 89, 97

1-100 $\Rightarrow$ 25

101-200 $\Rightarrow$ 21

7) Composite No $\rightarrow$ (other than prime)

4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21

even $\rightarrow$ Composite
odd $\rightarrow$ Composite

8) * 1 is Neither prime Nor Composite

* H.W (decimal HW Sheet):

$$1) \frac{(0.4)^3 \times (0.09)^2}{\sqrt{0.000729} \times \sqrt[3]{4.096}}$$

$$\frac{64 \times 81}{27 \times 16} = \frac{72}{24} = 3$$

$$\rightarrow \frac{3 \times 4}{3 \times 4} = \frac{12}{12} = 1$$

$$\frac{0.0012}{0.012} = 0.1$$

$$2) \frac{(1.2)^2 \times \sqrt{0.0256} \times (0.2)^2}{\sqrt[3]{0.512} \times \sqrt{0.0324} \times (0.02)^3}$$

$$\frac{144 \times 16 \times 4}{8 \times 18 \times 8}$$

$$= \underline{\underline{8}}$$

$$= \frac{2+2+2}{1+2+6}$$

$$= \frac{6}{9} = 3 \downarrow \Rightarrow \underline{\underline{8000}}$$

$$3) \frac{\sqrt{0.000064} \times \sqrt{0.000729}}{\sqrt[3]{0.000729} \times \sqrt[3]{0.000064}}$$

$$\frac{8 \times 27}{9 \times 4} = 6$$

$$\frac{3 \uparrow 3}{2+2} = \frac{6}{4} \Rightarrow \underline{\underline{0.06}}$$

$$4) \frac{(1.4)^2 \times (0.24)^2}{(0.12)^2 \times (0.7)^2}$$

$$\frac{196 \times 576}{144 \times 49} = 16$$

$$\frac{2+4}{6+2} = \frac{6}{8} = \underline{\underline{1600}}$$

$$5) \frac{(0.12)^3 \times (0.16)^3}{(0.8)^3 \times (0.04)^3}$$

$$\frac{144 \times 256}{512 \times 64} = 216$$

$$\frac{6+6}{3+6} = \frac{12}{9} \Rightarrow 3 \uparrow$$

$$= \underline{\underline{0.216}}$$

$$6) \frac{\sqrt{0.000576} \times \sqrt{0.0729}}{\sqrt{0.000324} \times \sqrt{0.000036}}$$

$$\frac{24 \times 27}{18 \times 6} = 6$$

$$= \frac{3+2}{3+3} = \frac{5}{6}$$

$$\Rightarrow \underline{\underline{60}}$$

$$7) \frac{\sqrt[3]{0.004096} \times \sqrt[3]{1.728}}{\sqrt[3]{0.000216} \times \sqrt[3]{0.000064}} = \frac{16 \times 12}{6 \times 4} = 8$$

$$\Rightarrow \frac{2+1}{2+2} = \frac{3}{4}$$

$$\Rightarrow \underline{\underline{80}}$$

$$8) \frac{\sqrt{0.0484} \times \sqrt{5.76} \times \sqrt{0.0676}}{\sqrt{0.04} \times (0.2)^3 \times \sqrt{1.21} \times \sqrt{0.09}} = \frac{22 \times 24 \times 26}{2 \times 8 \times 11 \times 3} = 13 \times 2 = 26$$

$$= \frac{2+1+2}{1+3+1+1} = \frac{5}{6} \Rightarrow 1 \downarrow$$

$$\Rightarrow \underline{\underline{13 \times 26}}$$

$$9) \frac{\sqrt{29.16} \times \sqrt{0.7056}}{\sqrt[3]{2744} \times \sqrt[3]{0.729}} = \frac{54 \times 24}{14 \times 9} = 36$$

$$\Rightarrow \frac{1+2}{1+1} = \frac{3}{2} \Rightarrow 1 \uparrow$$

$$\Rightarrow \underline{\underline{3.6}}$$

$$10) \frac{\sqrt{6.84} \times 1.024 \times 12.996}{(0.2)^3 \times \sqrt[3]{5.832} \times \sqrt{3.61} \times \sqrt[3]{2197}}$$

$$\Rightarrow \frac{4}{3+1+1+1}$$

$$\rightarrow \frac{4}{6}$$

$$\rightarrow 2 \downarrow$$

$$\Rightarrow \underline{\underline{800}}$$

$$= \frac{78 \times 32 \times 114}{8 \times 18 \times 13 \times 13}$$

$$= \frac{15 \times 114}{15 \times 13} = 2 \times 58$$

$$= 8$$

$$11) \frac{\sqrt[3]{2.097152} \times \sqrt[3]{0.373248}}{\sqrt{0.001296} \times \sqrt[3]{202.144}}$$

$$\rightarrow \frac{2+2}{3+1} = \frac{4}{4} \Rightarrow \underline{\underline{4}}$$

$$= \frac{128 \times 72}{36 \times 64}$$

exceptional case = $\underline{\underline{4}}$

$$12) \frac{\sqrt[3]{3.796416} \times 0.064 \times \sqrt[3]{592.704}}{(0.2)^2 \times \sqrt{0.49} \times \sqrt{0.16} \times \sqrt{15.21}}$$

$\frac{3+1}{2+1+1+1}$	$\frac{4}{156 \times 4 \times 84}$	$\frac{24}{196 \times 84^3}$
$= \frac{4}{5}$	$\frac{4 \times 7 \times 4 \times 39}{= 12}$	$\frac{4 \times 7 \times 4 \times 39}{= 24 \times 3}$
$\Rightarrow 1 \downarrow$		$= \frac{72}{39}$
$\Rightarrow \underline{\underline{12a}}$		$= 2$

Session-6

15/09/2026

* Decimal-2

$$1) \frac{\sqrt{6.4 \times 3.6 \times 0.0256}}{\sqrt[3]{4096 \times 0.8 \times 0.216}}$$

$$\frac{3}{2} = 1.5 \Rightarrow \underline{\underline{0.4}}$$

$$\frac{8 \times 6 \times 16}{16 \times 2 \times 6 \times 2} = 4$$

$$2) \frac{(0.4)^2 \times \sqrt{5.76 \times 6.4 \times 1.6 \times 1}}{\sqrt[3]{5.12 \times 17.28 \times 51.2 \times 0.8}}$$

$$\frac{16 \times 24 \times 8 \times 4}{8 \times 12 \times 8 \times 2} = 4 \times 2 = 8$$

$$= \frac{2+2}{2} = \frac{4}{2} \Rightarrow 2 \uparrow$$

0.08

158
141
14

133
117
11

excep library
↑ calc

3) $(1.8)^3 \times \sqrt{1.296 \times 0.144}$
~~Imp~~ $(0.06)^3 \times \sqrt[3]{7.29 \times 72.9}$

$$\frac{3+3}{6+1}$$

$$= \frac{6}{7} \Rightarrow 1 \downarrow$$

$$\Rightarrow 1440$$

5832 $\times$ 36 $\times$ 12
 216 $\times$ 9 $\times$ 9

Smart Work

$$\Rightarrow \overset{1}{18} \times \overset{2}{18} \times \overset{3}{18} \times \overset{4}{36} \times \overset{5}{12}$$

$$6 \times 6 \times 6 \times 9 \times 9$$

$$\Rightarrow 3 \times 4 \times 12$$

$$\Rightarrow 12 \times 12$$

$$\Rightarrow \underline{144}$$

4) $\sqrt{193.6 \times 1.296 \times 70.56}$
 $\sqrt[3]{2.744 \times 2.16 \times 0.1331}$

$$= \frac{3}{3}$$

$$= \underline{144}$$

$\overset{4}{44} \times \overset{6}{36} \times \overset{6}{84}$
 $14 \times 6 \times 11$
 $= 24 \times 6$
 $= 144^2$

5) $(2.4)^3 \times \sqrt{2.0736 \times 14.44}$
~~Imp~~ $\sqrt{3.61 \times 1.44} \times (0.012)^2 \times \sqrt[3]{1728}$

$$\Rightarrow \frac{3+3}{2+6+}$$

$$\rightarrow \frac{6}{8} \Rightarrow 2 \downarrow$$

$$\Rightarrow \underline{19200}$$

$\overset{2}{24} \times \overset{2}{24} \times \overset{2}{24} \times \overset{2}{144} \times \overset{2}{36}$
 $\overset{1}{19} \times \overset{1}{12} \times \overset{1}{144} \times \overset{1}{12}$
 $= \frac{24 \times 4 \times 36}{19}$
 $= \frac{96 \times 36}{19}$
 $= 24 \times 8$
 $= 192^3$

6) $(0.16)^3 \times \sqrt{3.136 \times 0.15876}$
~~Imp~~ $\sqrt[3]{27.44 \times 274.4} \times (0.8)^3$

$$= \frac{6+4}{1+3}$$

$$= \frac{10}{4} \Rightarrow 6 \uparrow$$

$$= \underline{0.000228}$$

$\overset{2}{4} \times \overset{2}{6} \times \overset{2}{15} \times \overset{4}{56} \times \overset{4}{128}$
 $14 \times 14 \times 8 \times 1 \times 8$
 $= \frac{8 \times 4 \times 128}{14}$
 $= 8 \times 4 \times 9$
 $= 36 \times 8$
 $= 228$

• speed:-

- 1) Tables
- 2) Factors
- 3) LCM
- 4) Fractions

• Factors : Imp No. 200 W.

38, 48, 54, 56, 60,
72, 80, 84, 90, 96,
105, 108, 112, 120,
126, 144, 150, 156,
168, 180, 192, etc.

eg- $60 = 2 \times 30$
 3×20
 4×15
 5×12
 6×10

5 pairs

1) $38 = 2 \times 19$

2) $48 = 2 \times 24$
 3×16
 4×12
 6×8

3) $54 = 2 \times 27$
 3×18
 6×9

4) $56 = 2 \times 28$
 4×14
 7×8

$$5) 60 = 2 \times 30$$

$$3 \times 20$$

$$4 \times 15$$

$$5 \times 12$$

$$6 \times 10$$

$$6) 72 = 2 \times 36$$

$$3 \times 24$$

$$4 \times 18$$

$$6 \times 12$$

$$8 \times 9$$

$$7) 80 = 2 \times 40$$

$$4 \times 20$$

$$5 \times 16$$

$$8 \times 10$$

$$8) 84 = 2 \times 42$$

$$3 \times 28$$

$$4 \times 21$$

$$6 \times 14$$

$$7 \times 12$$

$$9) 90 = 2 \times 45$$

$$3 \times 30$$

$$5 \times 18$$

$$6 \times 15$$

$$9 \times 10$$

$$10) 96 = 2 \times 48$$

$$3 \times 32$$

$$4 \times 24$$

$$6 \times 16$$

$$8 \times 12$$

$$11) 105 = 3 \times 35$$

$$5 \times 21$$

$$7 \times 15$$

$$12) 108 = 2 \times 54$$

$$3 \times 36$$

$$4 \times 27$$

$$6 \times 18$$

$$9 \times 12$$

$$13) 112 = 2 \times 56$$

$$4 \times 28$$

$$7 \times 16$$

$$8 \times 14$$

$$14) 120 = 2 \times 60$$

$$3 \times 40$$

$$4 \times 30$$

$$5 \times 24$$

$$6 \times 20$$

$$8 \times 15$$

$$10 \times 12$$

$$15) 126 = 2 \times 63$$

$$3 \times 42$$

$$6 \times 21$$

$$7 \times 18$$

$$9 \times 14$$

$$16) 144 = 2 \times 72$$

$$3 \times 48$$

$$4 \times 36$$

$$6 \times 24$$

$$8 \times 18$$

$$9 \times 16$$

$$12 \times 12$$

$$17) 150 = 2 \times 75$$

$$3 \times 50$$

$$5 \times 30$$

$$6 \times 25$$

$$10 \times 15$$

$$18) 156 = 2 \times 78$$

$$3 \times 52$$

$$4 \times 39$$

$$6 \times 26$$

$$12 \times 13$$

$$19) 168 = 2 \times 84$$

$$3 \times 56$$

$$4 \times 42$$

$$6 \times 28$$

$$7 \times 24$$

$$8 \times 21$$

$$12 \times 14$$

$$20) 180 = 2 \times 90$$

$$3 \times 60$$

$$4 \times 45$$

$$5 \times 36$$

$$6 \times 30$$

$$9 \times 20$$

$$10 \times 18$$

$$12 \times 15$$

$$21) 192 = 2 \times 96$$

$$3 \times 64$$

$$4 \times 48$$

$$6 \times 32$$

$$8 \times 24$$

$$12 \times 16$$